

Detectability-driven design of condition monitoring for three types of wind-turbine generators under identical winding faults

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Abstract

Whether the failures that matter in service will be observable is rarely a criterion when the architecture of an engineered system is chosen; condition monitoring is specified later. This paper treats diagnosability as a property of the triple (architecture, control structure, observation suite) and evaluates it with one scale-free statistic, the signal-to-drift ratio: the relative change of an observable feature under a failure mode of known extent, divided by the relative change the healthy system exhibits on its own. From it follow a minimum detectable extent and a detectable share per alternative, observation suite and failure-mode class, a decision table over the required extent and false-alarm level, and a six-step procedure that co-selects architecture and monitoring suite and tests whether a monitoring design transfers across a product family. The method is demonstrated on three wind-turbine generator topologies under identical winding short-circuits on open benchmark data. The same design-level fault produces different physical severities; where a fault becomes visible depends on the machine–drive combination, and the detectability difference between two machines on one bench comes from their healthy non-stationarity, not the signature; the cheapest suite depends on the alternative and on the requirement, voltage transducers lowering the permanent-magnet machine’s detectable extent and adding nothing for the induction machine; and a learned detector transfers across architectures only when its features are referenced to each trial’s own preceding healthy behaviour. Results carry bootstrap uncertainty, are limited to one machine per alternative, and are reproducible from the public data with the released code.

Keywords: design for diagnosability, architecture selection, sensor suite design, Design for X, condition monitoring, wind-turbine generators

1 Introduction

The architecture of an engineered system is chosen early, against criteria that are easy to quantify at that stage: performance, mass, cost, efficiency, manufacturability. Whether the failures that matter in service will be *observable*—and therefore detectable before they become expensive—is rarely among those criteria. Condition monitoring is specified later, by different people, for an architecture that is already fixed, and its difficulties are treated as signal-processing problems rather than as consequences of the architectural decision. The design community has methods that reason about failure at the concept stage—function–failure mapping and functional–failure reasoning evaluate architectures by the faults they can propagate (Tumer and Stone, 2003; Stone et al, 2005; Kurtoglu et al, 2010)—and the diagnosis community has structural methods that decide, from a model, which faults a given sensor set can detect (Krysander and Frisk, 2008; Blanke et al, 2016). Neither gives the designer an empirical *magnitude* of detectability that is calibrated against the healthy behaviour of each alternative and comparable across alternatives whose signals live in different physical units, and neither poses architecture and monitoring suite as one co-selection problem. This paper does both.

The central idea is to treat diagnosability as a property of the triple (architecture, control structure, observation suite) and to evaluate it with a single scale-free statistic computed from trials—measured or simulated—in which a failure mode of known extent is introduced into an otherwise healthy system. We call the statistic the signal-to-drift ratio (SDR): the relative change of an observable feature between a reference interval and a faulted interval, divided by the size of the relative changes the same feature exhibits between adjacent reference intervals when nothing is wrong. An SDR of one marks the point at which a fault of that extent becomes distinguishable from the ordinary non-stationarity of the healthy system at a chosen per-trial false-alarm rate. From the SDR follow the quantities a designer can write requirements on: the *minimum detectable extent* (MDE) of a failure-mode class for a given alternative and observation suite, and the *detectable share* (DS) of trials above a required extent in which it is detected. Both can be compared across alternatives and traded against the cost of the observation suite; a decision table over the requirement and the false-alarm level turns them into a selection.

The motivating domain is the generator of a large wind turbine. The choice between a permanent-magnet synchronous generator (PMSG), a squirrel-cage induction generator (SCIG), a wound-field synchronous generator (WFSG) and a doubly-fed induction generator is an established architecture-level trade (Polinder et al, 2006, 2013), and electrical-subsystem failures contribute substantially to offshore downtime (Carroll et al, 2016). Stator-winding insulation faults are the archetypal incipient failure: a short-circuit across a few turns with subtle signatures that evolves into a machine-destroying fault if undetected (Riera-Guasp et al, 2015). A rich literature detects them in a given machine (Thomson and Fenger, 2001; Nandi et al, 2005); almost none of it asks how the choice of machine changes the problem. A recent series of open benchmark datasets makes that question answerable empirically: Tominaga et al (2025a,b, 2024) subjected a PMSG, an SCIG and a variable-speed WFSG to the *same* inter-turn and inter-winding short-circuits, inserted at the same tapped positions of the stator

winding; the PMSG and SCIG share one bench and one protocol. The datasets were published for algorithm benchmarking and refrain from comparing the machines. We use them as the test-stage instantiation of a method whose object is the comparison itself.

The contributions are:

1. A definition of *alternative-dependent diagnosability* (Section 3.3) built on the SDR statistic, its calibration against a pooled no-fault null, a binomial reliability screen for observables, and three derived quantities: minimum detectable extent, detectable share above a required extent, and signature location.
2. A six-step procedure with default rules for its framing decisions, which co-selects an architecture and its observation suite under a detectability requirement through a decision table over the required extent and the false-alarm level, and which tests whether a monitoring design transfers across the alternatives of a product family.
3. A demonstration on three generator topologies under identical winding faults, with four findings that have design content: the design-level extent of a fault and its physical severity are different axes and must both be carried; the measurement group in which a fault becomes visible depends on the machine–drive combination; the cheapest adequate observation suite depends on the alternative *and* on the requirement; and a learned detector transferred across the machines only when its features were referenced to each trial’s own immediately preceding healthy behaviour, not to a commissioning baseline.
4. A sensitivity and uncertainty analysis of the tuning choices (null quantile, window length, screen), with trial-level and tap-pair-level bootstrap intervals, and open code that reproduces every result from the public data.

Section 2 positions the work; Section 3 defines the method; Section 4 describes the alternatives, the fault protocol, the observation suites and the computation; Section 5 reports the results; Section 6 draws out the design implications and the limits of what one machine per alternative supports; Section 7 concludes.

2 Background and related work

2.1 Failure and lifecycle properties at the concept stage

Systematic design methodology treats the evaluation of alternatives against lifecycle criteria as a core activity of the conceptual and embodiment phases (Pahl et al, 2007), and the Design-for-X programme has turned a growing list of such criteria into operational methods (Holt and Barnes, 2010). Reliability is the criterion closest to the present work. Failure mode and effects analysis remains the dominant design-stage instrument; recent work in this journal has re-founded it on design theory (Cabanes et al, 2021), and extended it with causal reasoning over failure data (Liu et al, 2026). What FMEA-type methods record about *detection* is an expert rating: the “D” column of a design FMEA (DFMEA) is assigned, not measured, and assigned for one architecture at a time.

The design community’s own line of failure reasoning at the concept stage runs from function–failure mapping (Tumer and Stone, 2003; Stone et al, 2005) to the

functional failure identification and propagation framework (FFIP), which evaluates conceptual architectures by simulating how faults propagate through their functional models (Kurtoglu and Tumer, 2008; Kurtoglu et al, 2010). These methods answer *which* failures an architecture can suffer and what they affect; they do not measure whether a failure of a given size would be visible to a given set of sensors, which is the question here. In the language of axiomatic design (Suh, 2001), the monitoring function and the control function of a converter-fed machine share the design parameter “current asymmetry”: the controller acts on the same quantity the monitor would like to observe, a coupling that the present method makes visible. Alternative selection is itself a methodological topic: Eaton et al (2026) show that technical measures omitted from a selection problem change which alternative is chosen. Diagnosability is, we argue, a routinely omitted technical measure, and because monitoring designs increasingly serve a family of variants (Simpson, 2004), whether a monitoring design transfers across alternatives is a family-level question. Following Blessing and Chakrabarti (2009), the demonstration in this paper is a support evaluation: it establishes that the method can be applied and what it yields, not that it improves design outcomes in practice.

2.2 Diagnosability, testability and sensor placement

Fault detection and isolation (FDI) has a mature theory of diagnosability for systems described by models. Structural analysis determines which faults can be detected and isolated from which sensor sets without numerical parameter values (Blanke et al, 2016; Krysanter and Frisk, 2008; Frisk et al, 2009), and has been applied to electric-vehicle drives with a diagnosability index that ranks candidate residual generators (Zhang and Rizzoni, 2015). In electronics, testability analysis answers the analogous question for test points (Simpson and Sheppard, 1994). Basseville (2001) gives statistical definitions of detectability and isolability for parametric models; the prognostics and health management (PHM) community has developed performance metrics (Saxena et al, 2008) and, closest to the SDR, metrics that grade candidate features by monotonicity, trendability and prognosability (Coble and Hines, 2013). Model-based approaches (Venkatasubramanian et al, 2003) require a model whose structure is known and trusted; data-driven diagnosis (Lei et al, 2020) requires data from the very system being designed, and its transfer across systems is an active problem (Pan and Yang, 2010).

Table 1 places the method among these. It does not require a model structure, only trials with a known fault extent; it does not require a trained detector, only observable features; and it produces a magnitude—the SDR—that is comparable across features, operating points and alternatives because it is normalised by each alternative’s own healthy non-stationarity rather than by a physical unit. Statistically it is the known-onset, fixed-sample mean-shift test of Basseville and Nikiforov (1993), expressed as a standardised effect size (Section 3.2); what is new is not the test but its use as a design measure over alternatives and suites, and the co-selection it supports. It deliberately stops at detectability: isolation and prognosis presuppose it.

Table 1 Positioning of the method among design-stage failure and diagnosability methods

Method		Input	Output	Sensing cost		Compares alternatives	alternatives
DFMEA rating	detection	expert judgement	ordinal D score	implicit		one at a time	
Function-failure mapping, FFIP (Stone et al, 2005; Kurtoglu et al, 2010)		functional model, failure knowledge	failure modes and propagation paths per concept	no		yes, qualitatively	
Structural diagnosability, sensor placement (Krysander and Frisk, 2008; Blanke et al, 2016)	diag-	model structure	detectable/isolable fault sets per sensor set	yes, sets	minimal	per model	
PHM feature suitability (Coble and Hines, 2013)	suit-	run-to-failure data	feature scores	no		no	
This paper		trials with known fault extent (measured or simulated)	MDE, detectable share and signature location per alternative and suite; decision table	yes, per suite		yes, quantitatively	quantitatively

2.3 Condition monitoring of electrical machines

Inter-turn short-circuits in stator windings are among the most studied incipient faults in electrical machines (Riera-Guasp et al, 2015). Motor-current signature analysis (Thomson and Fenger, 2001), negative-sequence current and impedance methods (Lee et al, 2003), transient models of turn faults (Tallam et al, 2002) and the review literature (Nandi et al, 2005) have established the physical signatures: an asymmetry of the winding produces negative-sequence components in the terminal quantities and a component at twice the fundamental frequency in the synchronous reference frame. Under closed-loop current control the picture changes: the current regulator acts against the negative-sequence current it can see, so the asymmetry migrates from the currents into the voltage commands and the controller states, with the degree of migration set by the current-loop bandwidth relative to twice the electrical frequency (Tallam et al, 2003; Cruz and Cardoso, 2004). In a wound-field synchronous machine the stator negative sequence additionally induces a component at twice the electrical frequency in the field current (Neti and Nandi, 2009), and in permanent-magnet machines with an accessible neutral the zero-sequence voltage is a sensitive indicator (Urresty et al, 2012). These facts are well known to drive specialists; they have not been turned into a measure that a design team can apply across candidate architectures. Machine-learning detectors, including our own (Canseven et al, 2026b; Sadik and Canseven, 2026), are typically developed for one machine; the transfer experiment in this paper is the design-level counterpart. Finite-element turn-fault models (Vaseghi et al, 2009) and digital twins (Tao et al, 2019) are the natural trial generators for a design-stage instantiation, and how such twins should be designed is an open question in this journal (Tipuric et al, 2026).

2.4 Generator topology as a wind-turbine design decision

The generator concept of a large wind turbine—direct-drive or geared; permanent-magnet, induction, wound-field or doubly-fed; full or partial power conversion—is selected early on cost, mass, efficiency and grid-compliance grounds (Polinder et al, 2006, 2013). Reliability enters the trade through failure-rate statistics of subassemblies (Carroll et al, 2016), and condition monitoring is added afterwards from a catalogue of commercial systems built around vibration, oil and electrical signals (Yang et al, 2014), not through the observability of the failures in each concept. Machine design optimisation has become highly effective at the electromagnetic and thermal objectives (Bramerdorfer et al, 2018); the present method adds an objective that has so far been missing from that toolbox.

3 Method: alternative-dependent diagnosability

3.1 Definitions and terms

Let $\mathcal{A}$ be a set of design *alternatives*. An alternative $a \in \mathcal{A}$ is an architecture together with the control structure it implies—for a generator, the machine topology and the converter control that the topology requires. Let $\mathcal{F}$ be a set of failure-mode classes; each class $f \in \mathcal{F}$ is parameterised by an *extent* $e \in E_f$ expressed at the design level, so that the same (f, e) can be realised in every alternative: for a winding fault the fraction of a winding branch that is short-circuited, for a bearing a defect size, for a structure a crack length. The design-level extent does not fix the physical severity of the event (Section 5.1), so the framing also names a *physical severity axis* on which the alternatives can be compared. Let $\mathcal{S}$ be a set of candidate *observation suites*; a suite $s \in \mathcal{S}$ is a set of measurements with a cost $c(s)$, and $\Phi(s)$ is the set of features computable from those measurements alone. Let $\mathcal{O}$ be the operating envelope, discretised into operating points o .

A *trial* is a record of the system in alternative a at operating point o containing a reference interval R in which the system is healthy and, immediately after it, a faulted interval F in which failure mode (f, e) is present. Trials may come from a physical test (test-stage instantiation) or from a validated simulation (design-stage instantiation, Section 3.5). We use these terms throughout: alternative (not topology or machine, except when a specific machine is meant), trial, observation suite, extent (design level) and severity (physical).

3.2 The signal-to-drift ratio

The reference interval is split into two adjacent halves R_1, R_2 , with R_2 adjacent to F . For a feature $\varphi \in \Phi(s)$ and a trial, let $\bar{\varphi}_{R_1}, \bar{\varphi}_{R_2}, \bar{\varphi}_F$ be the feature’s mean values over the intervals, and define the relative change under fault, referenced to the half adjacent to the fault, and the relative change between the two healthy halves,

$$\delta_\varphi = \frac{\bar{\varphi}_F - \bar{\varphi}_{R_2}}{|\bar{\varphi}_{R_2}|}, \quad \delta_\varphi^0 = \frac{\bar{\varphi}_{R_2} - \bar{\varphi}_{R_1}}{|\bar{\varphi}_{R_1}|}. \quad (1)$$

δ_φ^0 measures how much the feature moves when nothing is wrong: sensor noise, drift of the operating point, controller activity, thermal effects. Because every trial has a healthy reference interval, it is available from every trial of alternative a , faulted or not, and its absolute value pooled over the N_a trials forms an empirical null distribution of ordinary non-stationarity for that alternative and feature. The signal-to-drift ratio is

$$\text{SDR}_\varphi = \frac{|\delta_\varphi|}{q_\alpha(|\delta_\varphi^0|)}, \quad (2)$$

where q_α is the α -quantile of the pooled null; with $\alpha = 0.95$, $\text{SDR}_\varphi \geq 1$ means that the change under fault exceeds what the healthy system produces on its own in 95 % of comparable interval pairs.

What the SDR is statistically. For a roughly Gaussian null, $q_{0.95}(|\delta^0|) \approx 1.96 \sigma$, so $\text{SDR} \geq 1$ is a two-sided 5 % test on the shift of interval means with a known change time: the fixed-sample mean-shift test, i.e. the known-onset special case of the generalised likelihood ratio of [Basseville and Nikiforov \(1993\)](#). The SDR is therefore a trial-level standardised effect size (Cohen’s d on interval means, divided by 1.96), the detectable share defined below is an empirical power, and a share at a given α is one point of a receiver operating characteristic. Three assumptions are built in: the onset is known, the whole faulted interval is averaged, and the healthy reference lies immediately before the fault. The last makes the null one of sub-second non-stationarity; a reference established at commissioning has a wider null, and [Section 5.7](#) quantifies the difference. The SDR thus characterises the simplest detector on the feature—a threshold on the relative change of interval means—and the design quantities derived from it are statements about that detector; a detector that combines features or accumulates evidence over time can detect below $\text{SDR} = 1$, which is why the oracle variant and the transfer experiment of step 6 are part of the method.

Four properties make the SDR suitable as a design measure. It is *scale-free*: a current ratio, a voltage harmonic and a controller output are compared on one axis. It is *calibrated* against each alternative’s own non-stationarity, so a noisier alternative is not credited with detectability it does not have; the price is that numerator and denominator must be reported separately when alternatives are compared ([Section 5.2](#)), because a quieter null raises the SDR as much as a stronger signature does. It is *per trial*, so its distribution over trials carries the dependence on operating point and extent. And it requires *no detector*, so it can be computed before any algorithm is chosen. The relative form presumes a feature with a non-zero healthy level; for features whose healthy value is a noise floor an absolute-change variant with the same structure is used, and [Section 5.7](#) shows that the two rank the features alike.

Safeguards. (i) *Reliability screen.* On healthy trials—trials in which the “faulted” interval contains no fault, only the switching operation that would insert one—a feature should exceed $\text{SDR} = 1$ at the nominal rate $1 - \alpha$. A feature whose count of such false alarms in the n_h healthy trials is incompatible with that rate (one-sided binomial test at the 5 % level) is excluded from the design quantities for that alternative; with $n_h = 9$ and $\alpha = 0.95$ this means three or more false alarms. Features whose null quantile exceeds one (healthy relative change above 100 %) are excluded as uninformative.

(ii) *Fixed, nested and oracle feature*. A suite is evaluated by one *fixed* feature—the screened feature with the highest median SDR over the trials of a class, chosen once—or by the *oracle* feature, the best feature of each trial. The fixed variant is what a threshold detector achieves; the oracle variant is an upper bound for a detector that combines the suite’s features and, being a maximum over features, is reported with its false-alarm rate on healthy trials. Because the fixed feature is chosen on the trials it is evaluated on, a *nested* variant chooses it on all but one tap pair and evaluates it on the held-out pair. (iii) *Uncertainty*. Every design quantity carries a bootstrap interval: over tap pairs (the physical fault locations) for statements about a class, over operating points within an extent level for the minimum detectable extent, with Wilson intervals for shares.

3.3 Derived design quantities and the definition

From the SDR distribution over trials we define, for each alternative a , suite s and class f :

- the *diagnosability map* $D_a(s, f, e, o)$: the SDR of the suite’s fixed feature at extent e and operating point o ;
- the *minimum detectable extent* $\text{MDE}_a(s, f)$: the smallest tested extent level e^* such that $\text{median}_o D_a(s, f, e, o) \geq 1$ for every tested $e \geq e^*$ (limit-of-detection convention; equal extents at different winding locations form one level);
- the *detectable share* $\text{DS}_a(s, f \mid e \geq e_{\text{req}})$: the fraction of faulted trials with $e \geq e_{\text{req}}$ in which $\text{SDR} \geq 1$, written for a required extent so that it does not depend on how the test matrix samples the extent axis;
- the *signature location*: the measurement group—stator currents, terminal voltages, controller internals, mechanical quantities, excitation—whose best screened feature attains the highest SDR in the largest share of trials.

Alternative-dependent diagnosability is the map $(a, s, f) \mapsto (\text{MDE}_a, \text{DS}_a, \text{location})$. For a requirement $(f, e_{\text{req}}, \alpha)$ the *adequate* suites of alternative a are those with $\text{MDE}_a(s, f) \leq e_{\text{req}}$, and diagnosability is alternative-dependent for that requirement when the cheapest adequate suite differs between alternatives by more than the bootstrap uncertainty of the MDE. MDE and DS are the quantities against which requirements are written (“inter-turn faults of extent $\geq e_{\text{req}}$ shall be detectable over the operating envelope at a per-trial false-alarm rate $\leq 1 - \alpha$ ”); the signature location tells the designer *why* an alternative behaves as it does.

3.4 Design procedure

The procedure (Fig. 1) has six steps. Steps 3–4 are mechanical and fully specified by Eqs. (1)–(2); steps 1, 2, 5 and 6 are design decisions, for which the method supplies default rules.

1. **Frame**. Enumerate $\mathcal{A}$, the classes $\mathcal{F}$ with an extent scale E_f realisable in every alternative and a physical severity axis, the suites $\mathcal{S}$ and the envelope $\mathcal{O}$. Set e_{req} as the smallest extent whose growth to an unacceptable consequence is faster than the inspection interval, and $1 - \alpha$ as the per-trial false-alarm rate the operator tolerates

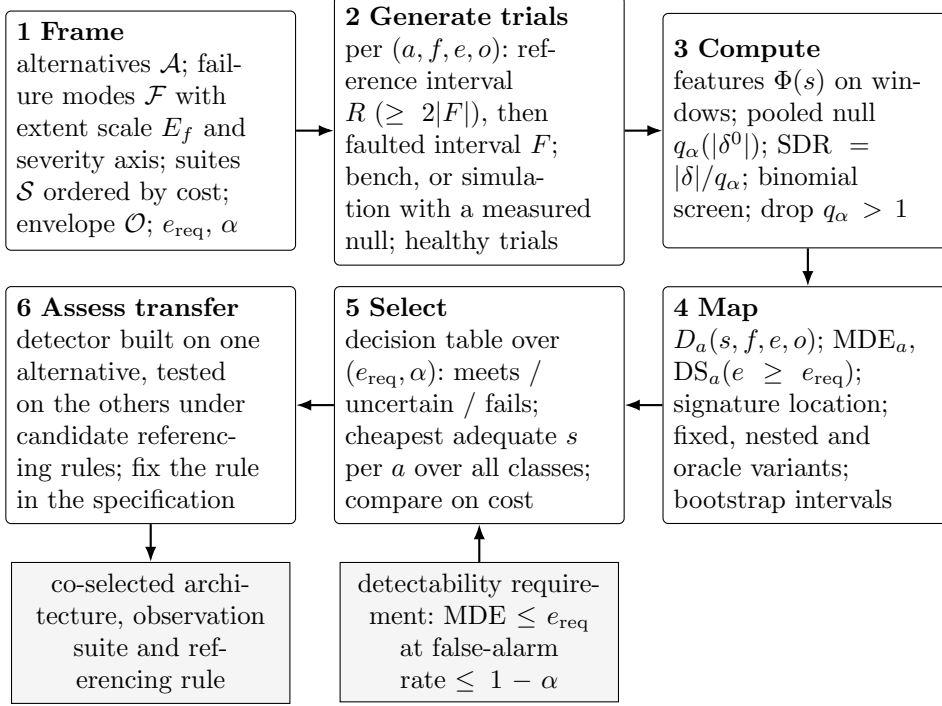


Fig. 1 The design procedure. Steps 3–4 are fully specified by Eqs. (1)–(2); steps 1, 2, 5 and 6 are design decisions for which Section 3.4 gives default rules

over that interval. Order the suites by cost: an ordinal rank by added hardware and integration by default, a monetary cost including installation and maintenance when available.

2. **Generate trials.** For each (a, f, e, o) obtain trials with a healthy reference interval of at least twice the faulted duration, immediately followed by the faulted interval; test at least the extent levels that bracket e_{req} and the corners of the envelope; include enough healthy trials with the same timing for the screen (nine allow the binomial rule above). At the test stage these are prototype or benchmark experiments; at the design stage the faulted numerator is simulated and the null is supplied as in Section 3.5.
3. **Compute.** Extract $\Phi(s)$ for all suites on sliding windows within the intervals, form the pooled null per alternative and feature, compute the SDR per trial, apply the screen.
4. **Map.** Build $D_a(s, f, e, o)$ and summarise it into MDE_a , DS_a and the signature location for the fixed, nested and oracle variants, with bootstrap intervals.
5. **Select.** Build the decision table: for each alternative, class and suite and for the candidate values of e_{req} and α , classify the suite as *meets* when the bootstrap probability that $\text{MDE} \leq e_{\text{req}}$ is at least 0.95, *fails* when it is at most 0.05, and *uncertain* otherwise. The adequate suite of an alternative is the cheapest that meets the requirement for *every* class in $\mathcal{F}$; an uncertain verdict is resolved by more trials

around e_{req} or by specifying the next suite as a fallback. The alternatives are then compared on the cost of their adequate suites together with the other selection criteria. The architecture decision is affected when, for some alternative, no affordable suite meets the requirement; otherwise the method changes the monitoring specification, not the architecture, and says so.

6. **Assess transfer.** If the suite and its detector must serve a family of alternatives, test whether a detector built on one alternative’s trials detects the others’ faults under the candidate feature-referencing rules (physical units; scaled on the target’s healthy trials; referenced to each trial’s own preceding healthy interval). The rule that transfers becomes part of the monitoring specification.

3.5 Design-stage instantiation

The demonstration in this paper is a test-stage instantiation on built machines. A design-stage instantiation must supply the two ingredients of the SDR separately. The numerator—the change of a feature under a fault of extent e —is what a finite-element or circuit model of each alternative with the fault injected can provide (Vaseghi et al, 2009). The denominator—the healthy non-stationarity of sensor noise, controller activity, thermal drift and operating-point wander—is a property of hardware and environment that such a model does not reproduce, and a simulated SDR that supplied its own null would be optimistic by an unknown factor. The null must therefore come from measured healthy records of hardware of the same class (a predecessor product, a prototype of one alternative, or a benchmark such as the one used here), transferred to the other alternatives on the assumption that the healthy non-stationarity of a feature depends on the sensing and control chain more than on the machine; that assumption is itself testable by comparing the null tables of the alternatives for which hardware exists (Online Resource 1, Table S10). A simulated instantiation is validated by comparing simulated and measured SDR on a few physical trials, the role the three benches would play in a design study of the next generator.

3.6 Scope

The method applies to failure-mode classes that can be seeded at a known extent, at a stationary operating point, with a healthy interval immediately before the fault, in every alternative; the extent scale must be realisable in all of them, and the physical severity axis is declared beside it. It evaluates detectability by a threshold on interval means, not the time to detection, and stops at detection: isolation, severity estimation and remaining-life prediction presuppose detectability and can be built on the same trials. It does not model the cost of missed detection, which enters through e_{req} and α . Run-to-failure data, in which the fault grows gradually and no healthy interval precedes a known onset, do not fit the trial structure and would need a different null.

Table 2 The three alternatives. PMSG and SCIG share one bench and one protocol; the WFSG bench differs where indicated. Ratings as given by the manufacturer (kVA for the synchronous machines, kW for the induction machine). 2L: two-level converter; 3L-NPC: three-level neutral-point-clamped converter; FOC: field-oriented control

	PMSG	SCIG	WFSG (variable speed)
Rating, poles	2.5 kVA, 4	2.5 kW, 4	2 kVA, 4 salient
Excitation	permanent magnets	induction (magnetising current)	wound field, static exciter
Converter and control	2L, FOC with encoder	2L, FOC with flux observer	3L-NPC, FOC with encoder
Acquisition	20 kHz, 3 s	20 kHz, 3 s	4 kHz, 1.3 s
Reference / faulted interval	0.68 s / 0.36 s	0.68 s / 0.36 s	0.49 s / 0.06–0.28 s
Fault impedance	2.6 Ω	2.6 Ω	1, 2.83, 5.66, 11.32 Ω
Operating points	3 speeds $\times$ 3 torques	same	716–1910 rpm, 4 torque levels (sparse)
Healthy trials	9	9	none
Fault trials	216	216	438 (198 at 2.83 Ω)

4 Demonstration: three generator topologies under identical winding faults

4.1 Alternatives and data

The three alternatives are laboratory-scale generators of the same power class on instrumented test benches at the University of São Paulo, whose records were published as open benchmark datasets (Table 2). The PMSG (Tominaga et al, 2025a) and the SCIG (Tominaga et al, 2025b) are four-pole, 230 V, double-star machines from the same manufacturer, rated 2.5 kVA and 2.5 kW respectively, installed on the *same* bench: the same induction-motor prime mover and frequency converter, the same rapid-control-prototyping platform and field-oriented control (FOC; rotor angle from an encoder for the PMSG, from a flux observer for the SCIG), the same sensors, the same acquisition (20 kHz, 3 s records), the same fault-insertion hardware and the same test script. The variable-speed WFSG (Tominaga et al, 2024) is a 2 kVA four-salient-pole machine on an earlier bench of the same group (Rocha et al, 2023) with a three-level converter, field-oriented control with encoder angle, 4 kHz acquisition and 1.3 s records. The PMSG–SCIG pair is therefore a controlled comparison in which everything but the machine and the control structure it requires is held fixed; the WFSG is a third alternative with the same fault protocol but a different bench, and is analysed with that difference kept explicit. Terminal voltages are measured at the machine terminals by each bench’s voltage transducers, whose bandwidth the data papers do not document, so the terminal-voltage features carry the sensing chain of their bench as well as the machine.

4.2 Failure-mode class and extent scale

All three machines were built with 24 tapped derivation points along their stator windings at the same relative positions, so that a short-circuit between two taps realises a fault of known extent (Online Resource 1, Table S9); the tap positions, given as

percentages of a winding branch from the neutral, are identical in the three data papers, which is what makes the extent scale realisable in every alternative. Two classes are used: inter-turn short-circuits (both taps on the same branch) and inter-winding short-circuits (taps on the two parallel branches of a phase). The extent e is the fraction of a branch between the two taps; it ranges from 2.7% to 12.1% for the twelve inter-turn cases and from 2.3% to 40.2% for the twelve inter-winding cases. Extents that coincide at different winding locations (2.8% twice, 7.4% three times, 11.6% twice) form one extent level. Faults were closed through a contactor and a series resistance after the machine had reached steady state: $2.6\,\Omega$ for 0.4s on the PMSG and SCIG; on the WFSG the experimenters raised the resistance with extent (1 Ω for the inter-turn cases below 11%, 2.83 Ω for the remaining inter-turn cases and the small inter-winding cases, 5.66 and 11.32 Ω for the large inter-winding cases), so that on that bench extent and impedance are confounded. The fault current is measured on all benches; it is used only as ground truth for the physical severity axis, never as a feature. Because the series resistance is of the order of the winding impedance of these small machines, the fault current is essentially the voltage induced in the shorted turns divided by the resistance, a *resistor-limited* severity that differs from a bolted fault, which is limited by the leakage reactance of the shorted turns (Section 5.1).

The PMSG and SCIG matrix is complete: 24 fault cases $\times$ 3 speeds (1200, 1500, 1800 rpm) $\times$ 3 torques (5.2, 6.4, 8.0 N m), the torque–speed pairs scaled from the operating region of the International Energy Agency (IEA) 15 MW reference turbine (Gaertner et al, 2020) to the laboratory speed range (the electrical frequency is not scaled), plus the healthy condition at each operating point; the SCIG generates with slip, at 59.6 Hz at 1800 rpm. The WFSG matrix covers the same 24 tap pairs at six speeds (716–1910 rpm) and four per-unit torque levels, sparsely and with two to six repetitions, so its operating points are not matched to the other two. The WFSG benchmark also contains phase-to-phase faults, a class absent from the PMSG/SCIG protocol; they are excluded from every analysis. The analyses use the WFSG’s 438 inter-turn and inter-winding trials at all impedances for the null, the feature ranking and the detectable shares, with the impedance as a covariate, and the 198 trials at 2.83 Ω —the impedance closest to the PMSG/SCIG value, covering four inter-turn tap pairs of 11.6–12.1% and four inter-winding pairs of 2.3–10.6%—for the minimum detectable extent and the decision table. Each table and figure states which set it uses.

4.3 Observation suites and features

Five observation suites are considered, ordered by the data access, hardware and integration they add to a bare generator–converter set (Table 3). *Drive-internal* uses only quantities the converter controller already has: the dq currents, its own dq voltage commands, the proportional–integral controller outputs, the modulation duty cycles, the DC-link voltage, the estimated electrical torque and the speed. $3\,CT$ (three current transducers) uses the three phase currents—from dedicated transducers or from the converter’s own current sensors routed to a monitoring system—with angle-free features only; for a converter-fed machine its cost is data access and bandwidth rather than new hardware. $3\,CT + 3\,VT$ adds three terminal voltage transducers (VT); $3\,CT + 3\,VT + drive$ combines the external transducers with controller access, which also

Table 3 Observation suites and the features they can compute. f_e : electrical frequency; “ $2f_e$ ”: amplitude of the component at twice the electrical frequency; I_1, I_2 (V_1, V_2): positive- and negative-sequence magnitudes at f_e ; THD: total harmonic distortion; PI: proportional–integral

Suite	Added access / hardware	Features
Drive-internal	controller data	I_d, I_q : std, $1f_e, 2f_e$; $V_d^{\text{cmd}}, V_q^{\text{cmd}}$: $2f_e$; PI outputs d, q : std, $2f_e$; duty-cycle negative sequence D_2/D_1 ; T_e : std, $1f_e, 2f_e$; DC-link std; speed std
3 CT	three phase currents, no angle	I_2/I_1 , RMS unbalance, harmonics 3, 5, 7 of I_a relative to the fundamental, THD
3 CT + 3 VT	+ 3 voltage transducers	3 CT features + V_2/V_1
3 CT + 3 VT + drive	+ controller data	all of the above + measured V_d, V_q : $2f_e$
+ torque transducer	+ shaft torque sensor	+ T_m : std, $1f_e, 2f_e$
Excitation (WFSG)	field-current measurement	I_f : std, $2f_e$

allows the measured terminal voltages to be transformed into the synchronous frame; and + *torque transducer* adds a shaft torque sensor. For the WFSG the field current is an additional drive-internal observable (the *excitation* suite), available because its exciter is static; a brushless exciter would not expose it (Neti and Nandi, 2009). Each feature is assigned to the cheapest suite whose measurements suffice to compute it. The suites are those the datasets can populate (zero-sequence voltage, stray flux, winding temperature, vibration and partial discharge are not recorded, and the angle-free Park’s-vector features were not computed), so the suite conclusions hold among the suites and features considered; the zero-sequence current is not used because the star points are isolated.

Features are computed on sliding windows of n_c electrical cycles with a hop of one cycle. The electrical frequency and the phase rotation are estimated per window from the phase slope of the current space vector (falling back to the nominal value outside 0.5–1.5 times it), so that no angle sensor is needed for the external suites; sequence components are obtained by Hann-windowed projection of each phase quantity onto the fundamental and symmetrical-component transformation; $1f_e$ and $2f_e$ amplitudes of synchronous-frame quantities by projection after mean removal. On the WFSG bench the measured converter-side phase voltages serve as terminal voltages, the synchronous-frame voltages are obtained from them with the encoder angle, the commanded voltages are the controller’s, and the electrical torque is computed from power and speed. Online Resource 1 (Section S1) lists the definitions.

4.4 Computation

For the PMSG and SCIG the reference interval is the 0.68 s before the contactor command, split into R_1 (0.32 s) and R_2 (0.36 s), and the faulted interval runs from 30 ms after the command to 10 ms before its release; because the relay picks up 85–90 ms after the command, that interval contains about 55 ms of healthy time, which dilutes the fault-interval mean and makes the PMSG/SCIG SDRs slightly conservative. For the WFSG the fault interval is located from the measured fault current and the reference interval, 0.49 s, is split at 0.30 s. The main PMSG–SCIG comparison uses

$n_c = 5$; the three-alternative comparison uses $n_c = 3$ for all three machines because the WFSG fault intervals are short: at 23.9–63.7 Hz a faulted interval holds between zero and eleven three-cycle windows, four trials hold none and are dropped, and 81 of 438 hold fewer than three. The sensitivity of every conclusion to $n_c \in \{3, 5, 8\}$ and to $\alpha \in \{0.90, 0.95, 0.99\}$ is reported, with the screen applied at each setting. The null is pooled over all 225 records of each of the PMSG and SCIG and over the 438 WFSG records, every record contributing a healthy reference interval. Uncertainty is quantified by bootstrap (Efron and Tibshirani, 1993) with 2000 resamples: the median SDR and the detectable share of a class are resampled over the twelve tap pairs, the same pairs for both machines so that the SCIG/PMSG ratio is matched; the minimum detectable extent is resampled over the operating-point trials within each extent level; shares also carry Wilson intervals; the null quantile is held at its full-sample value. The decision table uses 1000 resamples per suite, alternative, class and α , with $e_{\text{req}} \in \{3, 5, 8, 12\} \%$. All interval boundaries, thresholds, seeds and model settings are listed in Online Resource 1 (Section S2).

Step 6 of the procedure is instantiated as a window-level change detector: a gradient-boosted decision-tree (GBDT) classifier (Friedman, 2001) and, for the PMSG–SCIG pair, a logistic regression, trained to separate faulted windows from healthy ones. Within an alternative the detector is evaluated by five-fold cross-validation grouped by tap pair (healthy trials grouped by speed, so that every fold contains some), so that the tested fault locations are never seen in training; across alternatives it is trained on all trials of one machine and tested on all trials of another. Three feature-referencing rules are compared: physical units; features z -scored on the target machine’s healthy trials (a commissioning baseline); and the absolute z -score of each window against the mean and standard deviation of the *first half* of its own trial’s reference interval. Negatives are the reference and recovery windows of all trials plus the fault-interval windows of the healthy trials, the only negatives that contain the contactor operation without a fault. Because the windows used to normalise a trial are trivially separable under the third rule, every metric is reported on all windows and on the out-of-reference windows, and the false-alarm rate of the threshold that gives 1 % false positives on all negatives is reported on the healthy trials’ fault-time windows. The true-positive rate at 1 % false-positive rate is a point of the receiver operating characteristic, not an operating threshold.

All computations are implemented in Python and released with the paper; the pipeline runs from the feature caches to every table and figure in under thirty minutes on a four-core machine. A large language model was used as an assistant in writing the analysis code and in drafting and editing the text; every analysis and every reference was verified by the author (see the declarations).

5 Results

5.1 The same design-level fault is not the same physical fault

For the same tap pair the SCIG’s fault current is about 10 % lower than the PMSG’s (medians 3.3 against 3.7 A for inter-turn and 7.7 against 8.5 A for inter-winding pairs; Online Resource 1, Fig. S1), but the difference that matters for detection is relative to

Table 4 Decomposition of the SCIG/PMSG ratio of median SDR into its numerator (ratio of median $|\delta|$, the relative change under fault) and its denominator (ratio $q_{0.95}^{\text{PMSG}}/q_{0.95}^{\text{SCIG}}$ of the null quantiles); five-cycle windows

Class	Feature	$ \delta $ P	$ \delta $ S	q_{95} P	q_{95} S	Num.	Den.	SDR ratio
inter-turn	V_2/V_1	0.257	0.136	0.086	0.033	0.53	2.64	1.40
inter-turn	I_2/I_1	0.238	0.152	0.162	0.034	0.64	4.82	3.08
inter-turn	$I_d 2f_e$	0.291	0.107	0.563	0.016	0.37	35.09	12.95
inter-turn	$PI_d 2f_e$	0.289	0.107	0.557	0.016	0.37	34.71	12.91
inter-turn	$V_q 2f_e$	0.222	0.155	0.165	0.060	0.70	2.73	1.91
inter-turn	$V_q^{\text{cmd}} 2f_e$	0.197	0.139	0.162	0.039	0.71	4.17	2.94
inter-turn	D_2/D_1	0.251	0.111	0.090	0.010	0.44	8.96	3.98
inter-winding	V_2/V_1	5.318	1.569	0.086	0.033	0.30	2.64	0.78
inter-winding	I_2/I_1	1.833	1.869	0.162	0.034	1.02	4.82	4.91
inter-winding	$I_d 2f_e$	2.335	1.188	0.563	0.016	0.51	35.09	17.85
inter-winding	$PI_d 2f_e$	2.344	1.188	0.557	0.016	0.51	34.71	17.60
inter-winding	$V_q 2f_e$	3.688	1.495	0.165	0.060	0.41	2.73	1.11
inter-winding	$V_q^{\text{cmd}} 2f_e$	0.845	1.184	0.162	0.039	1.40	4.17	5.84
inter-winding	D_2/D_1	2.942	0.854	0.090	0.010	0.29	8.96	2.60

what the stator sensors see: the fault current is 1.45 times the pre-fault stator current (RMS to RMS, median over pairs) in the PMSG and 0.63 times in the SCIG for inter-turn faults, and 3.3 against 1.5 times for inter-winding faults, because the SCIG’s stator current carries the magnetising component (5.3 against 2.6 A RMS before the fault). The WFSG trials at the matching impedance give 1.20 (inter-turn, 11.6–12.1 %) and 1.03 (inter-winding, 2.3–10.6 %), and 0.65 at the lower impedance used for its small inter-turn extents. With a series resistance of the order of the winding impedance these currents are resistor-limited (the voltage induced in the shorted turns divided by the resistance), so they scale with volts per turn and speed; a bolted fault would be limited by the leakage reactance of the shorted turns, reach several times rated current and—the first-order difference between the topologies—could be de-excited in the SCIG and the WFSG but not in the PMSG. A fault specified as “ $e\%$ of a winding” therefore arrives at the sensors of the three alternatives as three different physical events; Section 5.3 uses both axes.

5.2 Where the signature appears, and why the SDR differs

Table S10 of Online Resource 1 gives the null of the SDR for the main features. It differs by up to a factor of thirty between the PMSG and the SCIG, and not only in sensed quantities: the null of the duty-cycle negative sequence, a controller output that involves no sensor, is five times larger on the PMSG, so the healthy non-stationarity is a property of the machine–drive combination, not of the transducers alone. Table 4 decomposes the SCIG/PMSG ratio of median SDR: for inter-turn faults the PMSG’s relative changes are the *larger* ones on every feature (SCIG/PMSG ratio of median $|\delta|$ 0.37–0.71), and the SCIG’s higher SDR is entirely a quieter null (ratio of null quantiles 2.6–35); for inter-winding faults the numerators are comparable (0.3–1.4) and the denominators again decide.

Fig. 2 shows the median SDR of every feature for the three alternatives, and Table S12 of Online Resource 1 the share of trials in which each measurement group carries the strongest signature. In the PMSG the terminal-voltage group is strongest in 67 % of inter-turn and 88 % of inter-winding trials, through V_2/V_1 (median SDR 3.0 and 62 at five-cycle windows; 76 and 94 % of trials detectable), and the stator-current negative sequence I_2/I_1 is weak for inter-turn faults (median SDR 1.5, 56 %). In the SCIG the controller group is strongest in 56 and 67 % of trials, through the $2f_e$ ripple of the d -axis current and its PI output (6.7 and 74; 79 and 100 %), the stator-current group in 28–33 % and the terminal-voltage group in 13 % and none. The two machines rank the 30 common features with a Spearman correlation of 0.70 (inter-turn) and 0.89 (inter-winding): the same features are informative in both, in a different order. In the WFSG the strongest inter-turn features are V_2/V_1 , the q -axis current ripple and the commanded q -axis voltage, and for inter-winding faults the q -axis ripple, the commanded voltage, the duty cycles and the third current harmonic; its field-current $2f_e$ ripple, an observable the other two machines lack, detected 86 % of the inter-turn trials at the matching impedance and all inter-winding trials (Online Resource 1, Table S1). The physical quantity behind the relocation is the incremental negative-sequence impedance $\Delta V_2/\Delta I_2$ seen at the terminals under inter-winding faults: 15 Ω in the PMSG, 7 Ω in the SCIG and 11 Ω in the WFSG (Online Resource 1, Table S7), i.e. the PMSG drive converts a given negative-sequence current disturbance into twice the terminal voltage that the SCIG drive does; for inter-turn faults the PMSG’s negative-sequence current increments are of mixed sign with a median indistinguishable from zero, which is why current-side features see little there. Both machines run the same field-oriented controller; the angle source (encoder against flux observer), the machine’s own negative-sequence impedance and the feature baselines differ, so the relocation belongs to the machine–drive combination, not to the control structure alone.

5.3 Minimum detectable extent

Table 5 gives the minimum detectable extent with its bootstrap interval, the detectable share with its Wilson interval and the median SDR with its tap-pair bootstrap interval for four representative features; Fig. 3 plots the SDR against the design-level extent (left) and the physical severity axis (right). Through the terminal-voltage negative sequence the PMSG detects inter-turn faults from 3.8 % [2.8, 3.8] and the SCIG from 7.4 % [7.4, 7.4]; through the stator-current negative sequence both need 7.4 %, and through the d -axis ripple the PMSG needs 12.1 % and the SCIG 7.4 % [3.8, 7.4]. The smallest tested inter-turn level (2.7–2.8 %) is not reliably detectable on either machine by any single feature (the SCIG’s V_2/V_1 and PI ripple exceed one at 2.7 % but not at 2.8 and 3.8 %). For inter-winding faults the SCIG detects from the smallest tested extent (2.3 %) on every feature except the q -axis voltage ripple, while the PMSG needs 2.3 % [2.3, 14.3] on V_2/V_1 but 9.7 % on I_2/I_1 and 21.7 % on the d -axis ripple. The SDR rises with extent in every machine (Spearman 0.51–0.95), and the MDE intervals are wide because the tested extents form three clusters (about 3, 7.5 and 12 % for inter-turn faults), so the SCIG/PMSG ratio of median SDR is the better basis for comparison: 3.1 [1.3, 5.6] for I_2/I_1 and 12 [2.6, 30] for the d -axis ripple on inter-turn faults, but 1.1 [0.4, 4.6] for V_2/V_1 and 1.5 [0.5, 3.2] for the q -axis voltage ripple. The

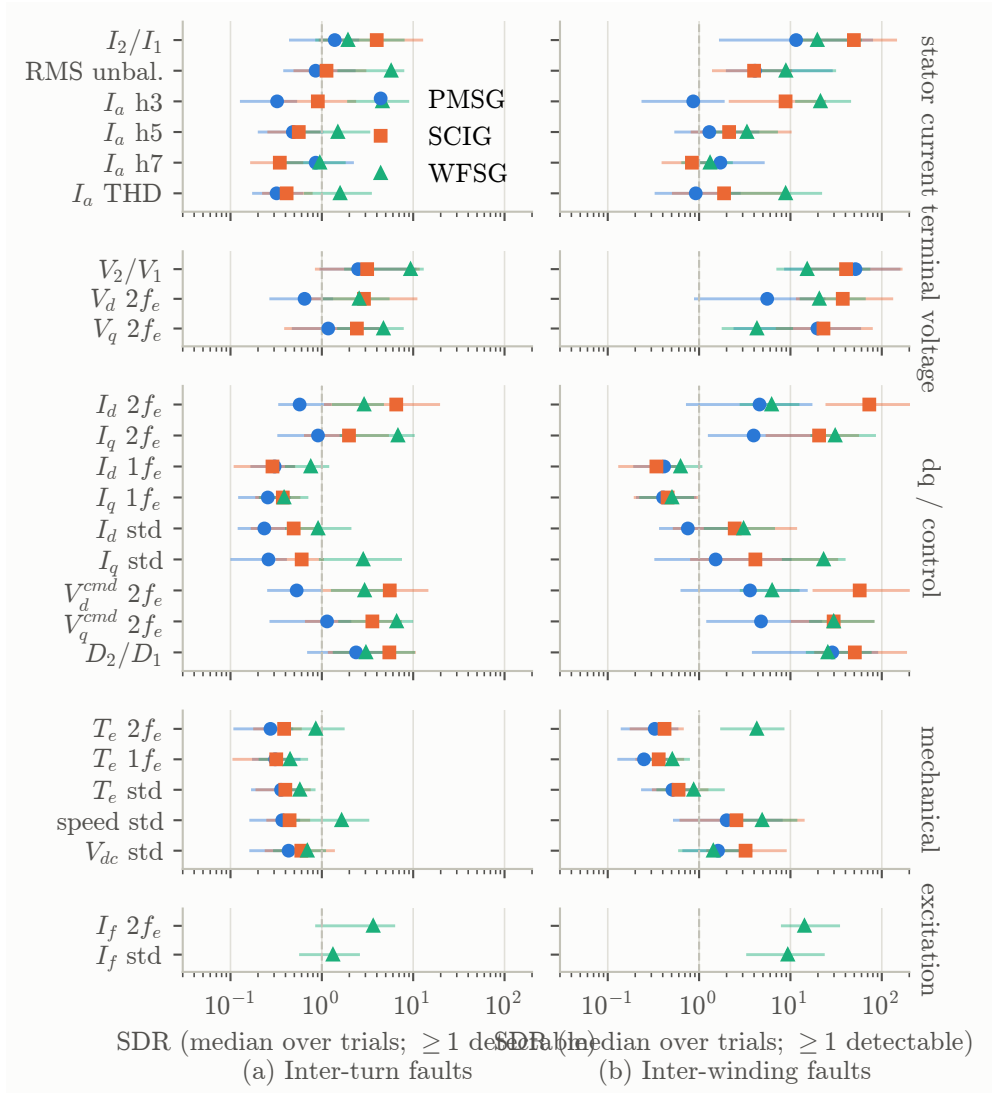


Fig. 2 Median signal-to-drift ratio of every feature for the three alternatives (dot) with interquartile range (bar), three-cycle windows; features to the right of the dashed line are detectable in more than half of the trials. The WFSG rows pool all its fault impedances

two machines differ where the signature relocates and are indistinguishable where it does not. On the physical axis (Fig. 3, right) the picture is the same and independent of the tap-pair design: at equal fault-to-stator current ratio the SCIG's current-side features lie above the WFSG's, which lie above the PMSG's, while on V_2/V_1 the three machines nearly coincide. For the WFSG, whose small inter-turn extents were tested only at the lower impedance, the design-level MDE at the matching impedance is bounded by the smallest extent tested there (11.6 %, detectable in 78–98 % of trials

Table 5 Minimum detectable extent (MDE, % of the winding branch between the taps; limit-of-detection convention; bootstrap 95 % interval over operating points within each extent level), detectable share (DS, %; Wilson 95 % interval), median SDR (95 % interval from a bootstrap over tap pairs, the same pairs for both machines) and the SCIG/PMSG ratio of medians, for four representative features; five-cycle windows, $\alpha = 0.95$. IT: inter-turn; IW: inter-winding; P: PMSG; S: SCIG. ‘-’: the largest tested extent is not detectable. Smallest tested extents: 2.7 % (inter-turn), 2.3 % (inter-winding)

Class	Feature	M.	MDE [CI]	DS [CI]	Median SDR [CI]	S/P ratio [CI]
IT	V_2/V_1	P	3.8 [2.8, 3.8]	76 [67, 83]	2.99 [1.21, 12.5]	1.1 [0.378, 4.56]
		S	7.4 [7.4, 7.4]	73 [64, 81]	4.17 [0.949, 15.5]	
IT	I_2/I_1	P	7.4 [7.4, 12.1]	56 [47, 65]	1.47 [0.601, 2.36]	3.07 [1.34, 5.63]
		S	7.4 [7.4, 7.4]	75 [66, 82]	4.52 [1.11, 11.7]	
IT	$PI_d 2f_e$	P	12.1 [11.6, 12.1]	31 [23, 40]	0.518 [0.389, 1.11]	11.8 [2.59, 28.8]
		S	7.4 [3.8, 7.4]	78 [69, 85]	6.69 [1.3, 18.3]	
IT	$V_q 2f_e$	P	7.4 [7.4, 11.6]	57 [48, 66]	1.35 [0.791, 4.58]	1.49 [0.464, 3.21]
		S	7.4 [7.4, 7.7]	60 [51, 69]	2.58 [0.516, 5.33]	
IW	V_2/V_1	P	2.3 [2.3, 14.3]	94 [88, 97]	61.9 [9.24, 181]	0.974 [0.696, 1.68]
		S	2.3 [2.3, 2.3]	100 [97, 100]	48.2 [15.3, 188]	
IW	I_2/I_1	P	9.7 [9.5, 21.7]	81 [73, 88]	11.3 [1.98, 64.3]	5.49 [1.75, 13.1]
		S	2.3 [2.3, 2.3]	100 [97, 100]	55.6 [12.5, 117]	
IW	$PI_d 2f_e$	P	21.7 [9.7, 21.7]	72 [63, 80]	4.21 [1.18, 13]	18.2 [12.3, 27.2]
		S	2.3 [2.3, 2.3]	100 [97, 100]	74 [22.9, 245]	
IW	$V_q 2f_e$	P	9.7 [2.3, 14.3]	90 [83, 94]	22.4 [3.24, 62]	1.13 [0.831, 1.9]
		S	9.7 [2.3, 9.7]	92 [85, 96]	24.8 [7.04, 78.6]	

depending on the suite); its 2.7–7.7 % inter-turn trials at 1Ω sit at $SDR \approx 1$ on V_2/V_1 and I_2/I_1 , and all its inter-winding extents from 2.3 % are detectable.

5.4 Operating envelope

Speed dominates the operating-point dependence (Online Resource 1, Table S3): in the PMSG the median SDR of the best feature and the detectable share rise from the lowest to the highest speed, in the SCIG less so and from a higher level; torque changes little. Two cautions apply: the dependence is consistent with a resistor-limited fault current that scales with speed, so it is a property of the emulation as much as of the machines; and the pooled null is a mixture over operating points (the share of trials whose healthy change exceeds the pooled 95th percentile is 0–16 % by speed for the key features), so the 5 % false-alarm reading holds on average over the envelope. Both make the lowest monitored speed a parameter of the requirement.

5.5 Observation suites and the design decision

Table 6 and Fig. 4 give the suite results for the PMSG and SCIG at five-cycle windows, Table S1 of Online Resource 1 for the three alternatives on the common feature set at three cycles, and Table 7 the decision table of step 5. For the PMSG, current transducers alone resolve inter-turn faults from 7.4 % and inter-winding faults from 9.7 %; three voltage transducers take these to 3.8 and 2.3 %, and neither controller access nor a torque transducer improves the fixed-feature result further (controller access raises the oracle share from 88 to 95 %). The drive-internal suite reaches 7.4 %

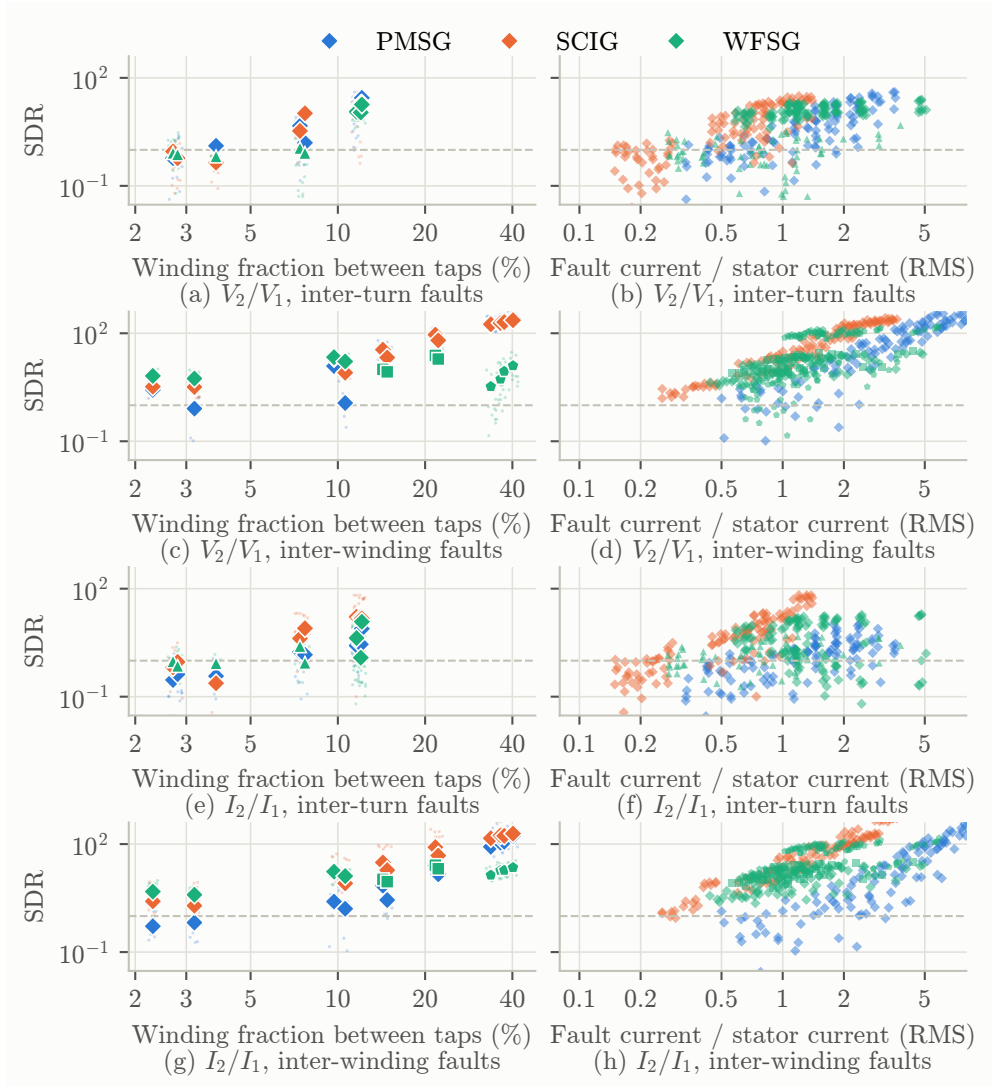


Fig. 3 Signal-to-drift ratio of V_2/V_1 (a–d) and I_2/I_1 (e–h) for the three alternatives, three-cycle windows, against the design-level fault extent (left column; small dots: trials; large markers: median per extent level and fault impedance, marker shape encoding the impedance) and against physical fault severity (right column; RMS fault current relative to the RMS pre-fault stator current, all trials)

for inter-turn faults through the duty-cycle negative sequence, a weaker version of the signature the measured terminal voltage carries. For the SCIG the cheapest suites already reach what any suite reaches—7.4% for inter-turn faults through I_2/I_1 or, with no added sensor, the d -axis PI output, and 2.3% for inter-winding faults—and additional hardware raises the detectable share rather than the reach. For the WFSG every suite detects all inter-winding faults from 2.3% and 78–98% of the inter-turn

Table 6 Observation suites versus what they detect, PMSG and SCIG (five-cycle windows): best fixed feature of the suite, its minimum detectable extent (MDE, %; limit-of-detection convention) and detectable share (DS, %; per-trial oracle share in brackets). Features failing the reliability screen (three or more false alarms on the nine healthy trials, or null quantile above one) are excluded for that machine; the oracle false-alarm rates are in Online Resource 1, Table S1

Suite	PMSG feature	MDE	DS [oracle]	SCIG feature	MDE	DS [oracle]
<i>Inter-turn faults</i>						
drive-internal	D_2/D_1	7.4	64 [87]	$PI_d 2f_e$	7.4	78 [99]
3 CT	I_2/I_1	7.4	56 [77]	I_2/I_1	7.4	75 [87]
3 CT + 3 VT	V_2/V_1	3.8	76 [88]	I_2/I_1	7.4	75 [90]
3 CT + 3 VT + drive	V_2/V_1	3.8	76 [95]	$PI_d 2f_e$	7.4	78 [99]
+ torque transducer	V_2/V_1	3.8	76 [95]	$PI_d 2f_e$	7.4	78 [99]
<i>Inter-winding faults</i>						
drive-internal	D_2/D_1	14.3	90 [100]	$PI_d 2f_e$	2.3	100 [100]
3 CT	I_2/I_1	9.7	81 [94]	I_2/I_1	2.3	100 [100]
3 CT + 3 VT	V_2/V_1	2.3	94 [97]	I_2/I_1	2.3	100 [100]
3 CT + 3 VT + drive	V_2/V_1	2.3	94 [100]	$PI_d 2f_e$	2.3	100 [100]
+ torque transducer	V_2/V_1	2.3	94 [100]	$PI_d 2f_e$	2.3	100 [100]

trials at the matching impedance, and the field current alone detects 86 and 100 %. The nested selection of the fixed feature changes no share by more than five points (Table S1); the oracle shares come with false-alarm rates of 44–78 % on the healthy trials for the large suites and are upper bounds only. A shaft torque transducer adds nothing for winding faults on the two machines on which it was available (the torque features’ healthy relative change exceeds 100 % on the SCIG and their median SDR is below 0.3 on the PMSG).

The decision table turns this into the selection of step 5. At $\alpha = 0.95$ no suite on any alternative meets an inter-turn requirement of 3 or 5 % with 95 % bootstrap confidence; the PMSG’s voltage suite is the only candidate below 5 % and its verdict is uncertain, a consequence of the three-cycle windows and the coarse extent grid. At 8 % the drive-internal suite meets the requirement on the PMSG and the SCIG, i.e. no added sensor is needed for either; the WFSG’s inter-turn verdicts are bounded by its smallest tested extent. For inter-winding faults the SCIG and the WFSG meet every requirement from 3 % with the drive-internal suite, whereas the PMSG needs current transducers at 12 % and is uncertain below (its five-cycle result, Table 6, reaches 2.3 % with voltage transducers). Relaxing α to 0.90 admits the PMSG’s voltage suite at 5 % for inter-turn faults; tightening it to 0.99 removes the PMSG’s current-only option at 12 % for inter-winding faults. The design content is therefore twofold: for a requirement around 8 % of a branch the three alternatives are equally diagnosable with drive-internal quantities; below 5 % only the PMSG with voltage transducers is a candidate and the evidence is not yet conclusive.

5.6 Transfer across alternatives

Table 8 reports step 6 for the PMSG–SCIG pair and Table S11 of Online Resource 1 for the three alternatives; the metrics on all windows and on the out-of-reference windows agree to within 0.02, so the windows used for normalisation do not drive the result. In physical units a gradient-boosted detector reaches an area under the

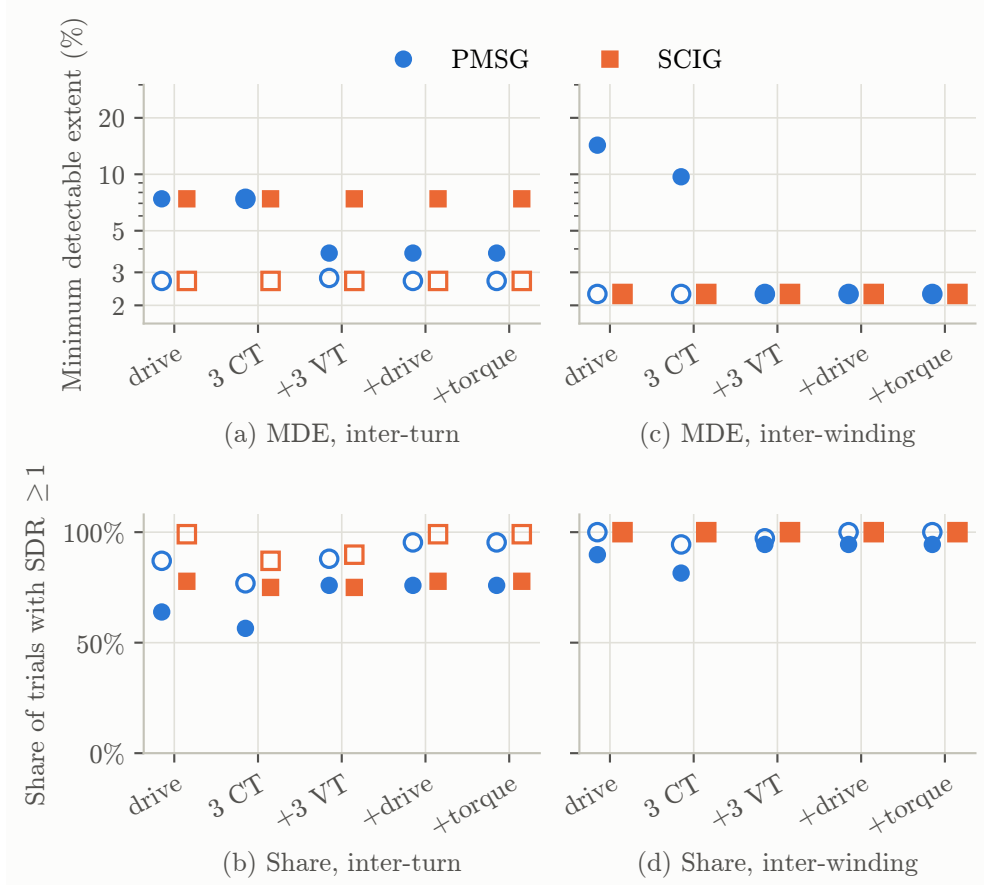


Fig. 4 Observation suites versus what they detect, PMSG and SCIG (five-cycle windows): minimum detectable extent (MDE; **a**, **c**) and detectable share (**b**, **d**) for the fixed best feature of each suite (filled; the feature is named in Table 6) and the per-trial oracle (hollow)

receiver operating characteristic curve (AUC) of 0.87–0.91 within a machine on unseen tap pairs but 0.78 (PMSG→SCIG) and 0.55 (SCIG→PMSG) across; with its features scaled on the target machine’s healthy trials, a commissioning baseline, 0.82 and 0.75, and a logistic detector under that rule falls below chance in one direction (0.45); with each feature referenced to the first half of the trial’s own reference interval, 0.91 and 0.84, against 0.91–0.94 within a machine, and a logistic detector transfers at 0.89 and 0.87. The threshold that gives 1% false positives on all negatives fires on 0–3% of the healthy trials’ fault-time windows, the only negatives that contain the contactor operation. Between the three alternatives, self-referenced detectors transfer in all six directions with AUC 0.80–0.98 on the out-of-reference windows, and a detector trained on two machines detects the faults of the third with 0.83 (PMSG), 0.88 (SCIG) and 0.97 (WFSG; optimistic, as that bench has no healthy trials). Restricted to each

Table 7 Decision table: cheapest observation suite whose minimum detectable extent meets the requirement e_{req} with bootstrap probability ≥ 0.95 , per alternative, fault class and false-alarm level α (three-cycle windows; suites ordered drive-internal $< 3\text{CT} < 3\text{CT}+3\text{VT} < 3\text{CT}+3\text{VT}+\text{drive}$; exc: WFSG field current). ‘(suite)?’: no suite meets with that confidence, the cheapest suite whose verdict is uncertain is shown; ‘none’: every suite fails. WFSG inter-turn verdicts are bounded by its smallest tested extent at the matching impedance (11.6%)

α	e_{req} (%)	PMSG		SCIG		WFSG	
		IT	IW	IT	IW	IT	IW
0.90	3	(3CT+3VT)?	(drive)?	none	drive	none	drive
0.90	5	3CT+3VT	(drive)?	none	drive	none	drive
0.90	8	drive	(drive)?	drive	drive	none	drive
0.90	12	drive	3CT	drive	drive	drive	drive
0.95	3	(3CT+3VT)?	(drive)?	none	drive	none	drive
0.95	5	(drive)?	(drive)?	none	drive	none	drive
0.95	8	drive	(drive)?	drive	drive	none	drive
0.95	12	drive	3CT	drive	drive	drive	drive
0.99	3	none	none	none	drive	none	drive
0.99	5	(3CT+3VT)?	none	none	drive	none	drive
0.99	8	drive	none	drive	drive	none	drive
0.99	12	drive	(drive)?	drive	drive	drive	drive

Table 8 Transfer of a window-level detector between PMSG and SCIG: AUC on all windows / AUC on the windows outside the normalising reference windows (second half of the reference interval, recovery windows and fault-time windows of the healthy trials). Within-machine columns use five-fold cross-validation grouped by tap pair (healthy trials grouped by speed); cross columns train on all trials of one machine and test on all trials of the other

Referencing	Model	P→P	S→S	P→S	S→P
physical units	logistic	0.78/0.78	0.79/0.80	0.71/0.71	0.59/0.59
physical units	GBDT	0.88/0.87	0.90/0.91	0.78/0.78	0.55/0.55
z on target’s healthy trials	logistic	0.78/0.78	0.79/0.80	0.71/0.71	0.44/0.45
z on target’s healthy trials	GBDT	0.88/0.87	0.90/0.91	0.82/0.82	0.75/0.75
$ z $ vs own reference	logistic	0.90/0.89	0.94/0.93	0.90/0.89	0.88/0.87
$ z $ vs own reference	GBDT	0.92/0.91	0.95/0.94	0.92/0.91	0.86/0.84

suite’s features (Online Resource 1, Table S6), every suite transfers at 0.82–0.95 when self-referenced and at 0.62–0.90 under the commissioning baseline.

5.7 Robustness and the two nulls

Table S4 of Online Resource 1 shows how the tuning choices move the design quantities, with the screen applied at every setting. The inter-turn MDEs of all five key features are the same at three, five and eight cycles on both machines and the detectable shares move by at most six points; for inter-winding faults one MDE moves (the PMSG’s V_2/V_1 , 9.7% at three cycles against 2.3% at five and eight). Tightening the null quantile from 0.95 to 0.99 lowers the shares by 3–9 points and moves two PMSG

inter-winding MDEs up one level; relaxing it to 0.90 raises the shares by up to 9 points and moves one down. No ordering of the two machines reverses, but the PMSG’s inter-winding suite conclusion depends on α and on the window length (Table 7). The screen removes between one and five features per machine and setting (Table S5); none of the five key features is removed at any setting, and the three-cycle result of V_2/V_1 on the PMSG (two false alarms in nine healthy trials) passes the binomial rule. An absolute-change variant of the SDR, which removes the healthy level of the feature from numerator and null, ranks the 30 features with a Spearman correlation of 0.92–0.98 against the relative variant and picks the same three top features in every machine and class. The bootstrap intervals (Table 5; Online Resource 1, Fig. S2) put the median SDRs within a factor of two to four and the shares within ± 8 –10 points.

Table S8 of Online Resource 1 quantifies the assumption that the healthy reference lies immediately before the fault. When the reference is instead the healthy trial recorded at the same operating point at another time—a commissioning baseline—the null widens by a factor of two to seven, the median SDR of the inter-turn features falls to about one (PMSG V_2/V_1 from 3.0 to 1.1; SCIG I_2/I_1 from 4.5 to 1.1) and their detectable shares to 44–59 %, while inter-winding faults remain detectable in 67–94 % of trials. The MDEs reported here are therefore statements about a short-horizon change detector; a monitor that compares against an old baseline needs the wider null.

6 Discussion

6.1 What the demonstration establishes about the method

The demonstration supports four claims about the method that do not depend on which machine came out ahead, and it corrects the naive form of two of them.

The design-level description of a failure mode does not fix its physical severity. A short-circuit of the same winding fraction produced a fault current that differed by about 10 % between the PMSG and the SCIG in amperes and by a factor of 2.3 relative to the stator current the sensors see; on the WFSG bench extent and fault impedance were confounded by the experimenters. A method that compares alternatives on a failure-mode class must therefore carry two severity axes—the design-level one, on which requirements are written, and a physical one, on which the alternatives are compared—and state which one a number refers to. For resistor-limited emulation the physical axis is predictable at the design stage from volts per turn, speed and the resistance; for a bolted fault from the leakage reactance of the shorted turns, and the topologies would then differ further in whether the fault can be de-excited.

Where a failure becomes observable is a property of the machine–drive combination, and the SDR must be read with its decomposition. In the PMSG the winding asymmetry appeared in the terminal voltage, in the SCIG in the magnetising-axis current and its controller action, in the WFSG additionally in the field current and the current harmonics; the incremental negative-sequence impedance under fault was twice as large in the PMSG drive as in the SCIG drive. The version of this claim in which “the control structure decides” is not supported: both machines run the same controller, and Table 4 shows that the SCIG’s higher SDR on current and d -axis features is entirely a quieter healthy null, its relative changes under fault being smaller than the

PMSG's. The SDR is designed to credit a quiet alternative, but a design team must know whether an alternative wins by signature or by silence, because the two generalise differently: a signature is a property of the fault physics, a null of the sensing and control chain and of the operating regime, which is exactly what a design-stage instantiation cannot simulate (Section 3.5).

The cheapest adequate observation suite depends on the alternative and on the requirement, and here it would change the monitoring specification, not the architecture. With three current transducers the PMSG reaches 7.4 % (inter-turn) and 9.7 % (inter-winding); three voltage transducers take it to 3.8 and 2.3 % and buy nothing on the SCIG, whose current-only or drive-internal suites already reach 7.4 and 2.3 %. The decision table makes the requirement explicit: for a required extent around 8 % of a branch all three alternatives meet it with drive-internal quantities and diagnosability does not discriminate between them; below 5 % only the PMSG with voltage transducers is a candidate, with an uncertain verdict at the confidence demanded. The method would change an architecture decision only if, for one alternative, no affordable suite met the requirement; here it changes what is specified for each. That is the honest answer to “so what for the designer”: a measured detection rating per alternative and suite, with its uncertainty, in place of an assigned one.

A monitoring design transfers across alternatives only under the right referencing rule, and the rule is a short-horizon one. A detector built on one machine and applied to another in physical units lost much of its performance; features scaled on the target machine's healthy trials—a commissioning baseline—recovered part of it and were harmful in one direction; features referenced to the first half of each trial's own reference interval transferred in all six directions between the three machines and from any two to the third. The between-trial null (Online Resource 1, Table S8) explains why: against a baseline recorded at another time, the inter-turn signatures sit at $\text{SDR} \approx 1$. For a product family this is a specification rule: the monitoring function should be a change detector on self-referenced deviations over a short horizon, and a monitor that compares against an old baseline must be calibrated on the wider null.

6.2 Design implications

For the wind-turbine generator, with the scope of Section 6.4: a PMSG with full-power conversion should be specified with terminal voltage measurement—or another observable of the terminal asymmetry, such as the zero-sequence voltage where a neutral is accessible (Urresty et al, 2012)—if inter-turn faults below about 7 % of a branch are to be detected by threshold monitoring, its controller's commanded quantities carrying a weaker version of the signature. An SCIG reaches the same detectability with current transducers alone or with no added sensor, and additional hardware buys robustness rather than reach. A wound-field machine with a static exciter offers a further drive-internal channel, the field current, whose $2f_e$ ripple detected every inter-winding fault and most inter-turn faults; a brushless exciter would not expose it (Neti and Nandi, 2009). A shaft torque transducer, as filtered on these benches, is not worth specifying for winding faults. Two industrial parameters change these statements and are computable at the concept stage: the ratio of current-loop bandwidth to twice the

electrical frequency, which for a direct-drive multi-megawatt PMSG at a fundamental near 10 Hz will push the signature further into the controller and the voltage than in these 40–60 Hz machines; and the grid-connected, unregulated stator of a doubly-fed generator, where the migration is absent and current-signature routes regain their reach. More generally, the method gives DFMEA’s detection rating a measured basis and digital twins a concrete design-stage role: the twin supplies the faulted numerator, measured healthy records supply the null, and a few physical trials validate the pair.

6.3 Applying the method elsewhere

Nothing in Section 3 is specific to electrical machines, but the scope conditions of Section 3.6 are. A gearbox family would use tooth-damage size as extent, vibration and oil-debris suites, and seeded-damage trials with a healthy run before each damage step; a battery-pack family would use cell-capacity loss, voltage, temperature and impedance suites, and trials in which a degraded cell is switched into a healthy pack. Run-to-failure records do not fit the trial structure without a different null; everything else carries over unchanged.

6.4 Threats to validity

The most important limitation is that each alternative is represented by one physical machine. The differences reported are differences between three machines; attributing them to topology rather than to the individual units requires replicated machines or simulated populations of each alternative under parameter variation, validated against the measured units. We regard the demonstration as establishing the method and the mechanisms, not the population-level ranking of the topologies, and the wording of the findings is scaled accordingly. The two benches also differ in converter, controller, sampling rate and sensing chain.

The fault emulation is resistor-limited and the benches are laboratory scale (2–2.5 kVA); the operating points were scaled from a 15 MW turbine but the electrical frequency and the fault physics were not. Only winding faults and only electrical and controller observables were studied, and the suites are those the datasets can populate: zero-sequence voltage, stray flux, temperature, vibration and partial discharge are absent, and the angle-free Park’s-vector features were not computed. On the WFSG the fault impedance is confounded with extent; only four tap pairs per class exist at the impedance matching the other two machines, so its inter-turn MDE is bounded, not measured; it has no healthy trials, so the reliability screen cannot be applied to it and its transfer entries lack nuisance-transient negatives; and 81 of its 438 trials hold fewer than three three-cycle windows. The PMSG/SCIG faulted interval contains about 55 ms of healthy time before the relay picks up, which dilutes those SDRs slightly relative to the WFSG’s, whose interval is located from the measured fault current.

Statistically, the pooled null is a mixture over operating points and holds its nominal false-alarm rate on average, not per speed; the null quantile is held fixed inside the bootstrap; the fixed feature is chosen on the trials it is evaluated on, which the nested variant shows to be immaterial here; the oracle variant is a maximum over features and its false-alarm rate is reported for that reason; the minimum detectable extent is

a discrete quantity on a coarse extent grid, so the continuous SDR and its ratio are the better basis for comparing alternatives. Other nuisance transients (load steps, grid events) were not in the data. Finally, the SDR characterises a short-horizon change detector; the between-trial null shows that a monitor referenced to an old baseline faces a null two to seven times wider, and every MDE in this paper should be read with that qualification.

7 Conclusions

We have proposed a method that makes the diagnosability of failure modes a criterion for choosing between architectural alternatives and, jointly, for choosing the observation suite that will monitor the chosen architecture. Its core is a scale-free, self-calibrated statistic, the signal-to-drift ratio, from which a minimum detectable extent and a detectable share above a required extent can be computed for every combination of alternative, suite and failure-mode class, with a reliability screen, bootstrap intervals and a decision table over the required extent and the false-alarm level; a six-step procedure with default rules for its framing decisions turns these into a co-selection and a transferability specification.

Applied to three wind-turbine generator topologies subjected to identical winding short-circuits on open benchmark data, the method showed that the same design-level fault is not the same physical fault across the machines; that where a fault becomes visible depends on the machine–drive combination, with the difference in detectability between the two machines on one bench arising from their healthy non-stationarity rather than from the strength of the signature; that the cheapest observation suite meeting a requirement depends on the alternative and on the requirement, terminal voltage transducers lowering the permanent-magnet machine’s detectable extent and adding nothing for the induction machine, while for a requirement of about 8 % of a branch all three alternatives are equally diagnosable with drive-internal quantities; and that a learned detector transfers across the three machines only when its features are referenced to each trial’s own preceding healthy behaviour, a commissioning baseline being insufficient for inter-turn faults. The findings are robust in their orderings to the method’s tuning, are reported with tap-pair-level uncertainty, and are limited by one physical unit per alternative, laboratory scale, resistor-limited fault emulation and a single failure-mode class.

Three extensions follow. First, the design-stage instantiation: simulated faulted trials combined with measured healthy nulls of hardware of the same class, validated on a few physical trials, would let the method be applied before hardware exists and turn machine-level differences into population-level statements. Second, other failure-mode classes and observables—bearing and rotor faults with vibration and flux observables, zero-sequence and Park’s-vector features for the winding faults studied here—would test the claim to generality within the domain, and a mechanical or electrochemical family across domains. Third, the cost side of the selection, treated here as an ordinal ranking of suites, deserves a model including integration, maintenance and the cost of missed detection, so that the co-selection can be posed as an optimisation. The released code reproduces every result from the public data and is the starting point for those extensions.

Acknowledgements. The author thanks the InnovaPower group of the University of São Paulo for publishing the benchmark datasets on which this work rests.

Declarations

Funding. No specific funding was received for this work.

Competing interests. The author declares no competing interests.

Ethics approval and consent. Not applicable; the work uses no human or animal subjects and no personal data.

Data availability. All data are public: the PMSG, SCIG and synchronous-generator datasets of the InnovaPower MitDev-Eletrica project (<https://github.com/InnovaPower/MitDev-Eletrica>; Zenodo DOIs 10.5281/zenodo.15741561, 10.5281/zenodo.17161986 and 10.5281/zenodo.13685630), licensed CC BY 4.0. The derived per-trial tables (features, signal-to-drift ratios, bootstrap samples) are provided with the code.

Code availability. The complete analysis pipeline (feature extraction, signal-to-drift computation, sensor-suite evaluation, transfer experiments, sensitivity analysis and all figures) is released under an open licence at <https://github.com/htcanseven/Claude> (directory `scripts/`; an archived release with a DOI will be deposited on Zenodo at acceptance); it reproduces every number in this paper from the public data.

Author contributions. H.T.C. conceived the method, designed and carried out the analysis, and wrote the paper.

Use of generative AI. A large language model (Claude, Anthropic) was used under the author’s direction as an assistant for code development, literature retrieval, data analysis and drafting and editing of the text. The author designed the study, verified all analyses, checked every reference against its published record, and takes full responsibility for the content. No AI-generated images are used; all figures are produced by the released code from the data.

Supplementary information. Online Resource 1 (PDF) contains the feature definitions, the computation details, the supplementary tables S1–S12 and the supplementary figures S1–S2 referred to in the text.

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